

```

161 -- 1. Find the average salesman's salary.
162 delimiter //
163 • create function average_salesman_salary()
164 returns decimal(15,4)
165 deterministic
166 begin
167     declare average decimal(15,4);
168     set average = (
169         select avg(salary)
170         from salesman
171     );
172     return average;
173 end //
174 delimiter ;
175
176 • select average_salesman_salary() as average_salesman_salary;
177

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	average_salesman_salary			
▶	17916.6667			

```

178 -- 2. Find the name of the highest paid salesman.
179 • drop function if exists highest_paid_salesman;
180 delimiter //
181 • create function highest_paid_salesman()
182 returns varchar(25)
183 deterministic
184 begin
185     declare name_salesman varchar(25);
186     set name_salesman = (
187         select salesman_name
188         from salesman
189         where salary = (
190             select max(salary)
191             from salesman
192         )
193     );
194     return name_salesman;
195 end //
196 delimiter ;
197
198 • select highest_paid_salesman() as highest_paid_salesman;
199

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	highest_paid_salesman			
▶	Phat			

```

200 -- 3. Find the name of the salesman who is paid the lowest salary.
201 delimiter //
202 • create function lowest_paid_salesman()
203 returns varchar(25)
204 deterministic
205 begin
206     declare name_salesman varchar(25);
207     set name_salesman = (
208         select salesman_name
209         from salesman
210         where salary = (
211             select min(salary)
212             from salesman
213         )
214     );
215     return name_salesman;
216 end //
217 delimiter ;
218
219 • select lowest_paid_salesman() as lowest_paid_salesman;
220

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

lowest_paid_salesman

► Deb

```

221 -- 4. Determine the total number of salespeople employed by the company
222 • drop function if exists total_number_salespeople;
223 delimiter //
224 • create function total_number_salespeople()
225 returns int
226 deterministic
227 begin
228     return (select count(salesman_number) from salesman);
229 end //
230 delimiter ;
231
232 • select total_number_salespeople() as total_number_salespeople;
233

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

total_number_salespeople

► 6

```

234    -- 5. Compute the total salary paid to the company's salesman
235    delimiter //
236 •   create function total_salary_salesman()
237     returns decimal(15,4)
238     deterministic
239     begin
240         return (select sum(salary) from salesman);
241     end //
242     delimiter ;
243
244 •   select total_salary_salesman() as total_salary_salesman;
245

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	total_salary_salesman			
▶	107500.0000			

```

246 -- 6. Find Clients in a Province
247 delimiter //
248 • create function find_province
249 (
250     in_province char(25)
251 )
252 returns int
253 deterministic
254 begin
255     return (
256         select count(province)
257         from clients
258         where province = in_province
259     );
260 end //
261 delimiter ;
262
263 • select find_province('Binh Duong') as BinhDuong;
264

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	BinhDuong			
▶	4			

```

265 -- 7. Calculate Total Order Amount
266 • drop function if exists total_order_amount;
267 delimiter //
268 • create function total_order_amount()
269 returns decimal(15,4)
270 deterministic
271 begin
272     return (
273         select sum(total_price) from (
274             select (sod.order_quantity * p.sell_price) total_price
275             from salesorderdetails sod
276             inner join product p using(product_number)
277         ) as cnt_list
278     );
279 end //
280 delimiter ;
281
282 • select total_order_amount() as total_order_amount;

```

783

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	total_order_amount
▶	101028.0000

```

392 -- a) Check constraint to value of gender in "Nam" or "Nu".
393 • alter table employees
394 add constraint check_gender_employee
395 check (gender in ('Nam', 'Nu'));
396
397 • alter table relative
398 add constraint check_gender_relative
399 check (gender in ('Nam', 'Nu'));
400
401 -- b) Check constraint to value of salary > 0.
402 • alter table employees
403 add constraint check_salary_employee
404 check (salary > 0);
405
406 -- c) Check constraint to value of relationship in Relative table in "Vo chong", "Con trai", "Con gai", "Me ruot", "Cha ruot".
407 • alter table relative
408 add constraint check_relationship_relative
409 check (relationship in ('Vo chong', 'Con trai', 'Con gai', 'Me ruot', 'Cha ruot'));

```

```

411 -- III. Writing SQL Queries by database "Assignment2"
412 -- a) Look for employees with salaries above 25,000 in room 4 or employees with salaries above 30,000 in room 5.
413 • select *
414 from employees
415 where (salary > 25000 and departmentID = 4) or (salary > 30000 and departmentID = 5);
416

```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	employeeID	firstName	middleName	lastName	dateOfBirth	gender	salary	address	managerID	departmentID
▶	333	Tung	Thanh	Nguyen	1945-08-12	Nam	40000	638 Nguyen Van Cu, Q5, TPHCM	888	5
	666	Hung	Manh	Nguyen	1952-09-15	Nam	38000	975 Le Lai, P3, Vung Tau	333	5
	987	Nhan	Thi	Le	1931-06-20	Nu	43000	291 Ho Van Hue, Q.PN, TPHCM	888	4
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

```

417 -- b) Provide full names of employees in HCM city.
418 • select firstName, middleName, lastName
419 from employees
420 where address like '%TPHCM%';
421

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	firstName	middleName	lastName
▶	Tien	Ba	Dinh
	Tung	Thanh	Nguyen
	Nhan	Thi	Le

```

422 -- c) Indicate the date of birth and address of Dinh Ba Tien staff
423 • select dateOfBirth
424 from employees
425 where firstName = 'Tien' and middleName = 'Ba' and lastName = 'Dinh';
426

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	dateOfBirth
	1955-09-01

```

427 -- d) The names of the employees of Room 5 are involved in the "San pham X" project and this employee is directly managed by "Nguyen Thanh Tung"
428 • select e.firstName, e.middleName, e.lastName
429 from employees e
430 inner join projects p using(departmentID)
431 where p.projectName = 'San Pham X' and e.departmentID = 5 and e.managerID in (
432     select e2.employeeID from employees e2
433     where e2.firstName = 'Tung' and e2.middleName = 'Thanh' and e2.lastName = 'Nguyen'
434 );
435

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

firstName	middleName	lastName
▶ Hung	Manh	Nguyen
Tam	Thanh	Tran
Tien	Ba	Dinh

```

436 -- e) Find the names of department heads of each department.
437 • select e.firstName, e.middleName, e.lastName
438 from employees e
439 where e.employeeID in (
440     select managerID from department
441 );

```

442

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	firstName	middleName	lastName
▶	Quyen	Ngoc	Vuong
	Quang	Hong	Tran
	Tung	Thanh	Nguyen

```

443 -- f) Find projectID, projectName, projectAddress, departmentID, departmentName, departmentID, dateOfEmployment
444 • select projectID, projectName, projectAddress, departmentID, departmentName, departmentID, dateOfEmployment
445 from projects
446 inner join department using(departmentID);

```

447

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	projectID	projectName	projectAddress	departmentID	departmentName	departmentID	dateOfEmployment
▶	1	San pham X	VUNG TAU	5	Nghien cuu	5	1978-05-22
	2	San pham Y	NHA TRANG	5	Nghien cuu	5	1978-05-22
	3	San pham Z	TP HCM	5	Nghien cuu	5	1978-05-22
	10	Tin hoc hoa	HA NOI	4	Dieu hanh	4	1985-01-01
	20	Cap Quang	TP HCM	1	Quan ly	1	1971-06-19
	30	Dao tao	HA NOI	4	Dieu hanh	4	1985-01-01

```

448 -- g) Find the names of female employees and their relatives
449 • select concat_ws(' ', e.lastName, e.middleName, e.firstName) employeeFullName, r.relativeName
450 from employees e
451 left join relative r using(employeeID)
452 where e.gender = 'Nu';

```

453

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	employeeFullName	relativeName
▶	Vuong Ngoc Quyen	NULL
	Le Thi Nhan	Dang

```

454 -- h) For all projects in "Hanoi", list the project code (projectID), the code of the project lead department (departmentID),
455 -- the full name of the manager as well as the address (Address) and date of birth (dateOfBirth) of the Employees.
456 • select p.projectID, d.departmentID, concat_ws(' ', e.lastName, e.middleName, e.firstName) managerName, e.address, e.dateOfBirth
457 from departmentaddress da
458 join projects p using(departmentID)
459 join department d using(departmentID)
460 join employees e using(departmentID)
461 where da.address = 'HA NOI';

```

462

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	projectID	departmentID	managerName	address	dateOfBirth
▶	30	4	Tran Hong Quang	980 Le Hong Phong, Vung Tau	1959-03-29
	10	4	Tran Hong Quang	980 Le Hong Phong, Vung Tau	1959-03-29
	30	4	Le Thi Nhan	291 Ho Van Hue, Q.PN, TPHCM	1931-06-20
	10	4	Le Thi Nhan	291 Ho Van Hue, Q.PN, TPHCM	1931-06-20
	30	4	Bui Thuy Vu	332 Nguyen Thai Hoc, Quy Nhon	1958-07-19
	10	4	Bui Thuy Vu	332 Nguyen Thai Hoc, Quy Nhon	1958-07-19

```

463 -- i) For each employee, include the employee's full name and the employee's line manager
464 • select concat_ws(' ', e.lastName, e.middleName, e.firstName) employeeName,
465         concat_ws(' ', m.lastName, m.middleName, m.firstName) managerName
466 from employees e
467 join employees m on e.managerID = m.employeeID;
468

```

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	employeeName	managerName
▶	Dinh Ba Tien	Nguyen Thanh Tung
	Nguyen Thanh Tung	Vuong Ngoc Quyen
	Tran Thanh Tam	Nguyen Thanh Tung
	Nguyen Manh Hung	Nguyen Thanh Tung
	Tran Hong Quang	Le Thi Nhan
	Le Thi Nhan	Vuong Ngoc Quyen
	Bui Thuy Vu	Le Thi Nhan

```

469 -- j) For each employee, indicate the employee's full name and the full name of the head of the department in which the employee works.
470 • select concat_ws(' ', e.lastName, e.middleName, e.firstName) employeeName,
471         concat_ws(' ', m.lastName, m.middleName, m.firstName) managerName
472 from employees e
473 join department d using(departmentID)
474 join employees m on d.managerID = m.employeeID;

```

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	employeeName	managerName
▶	Dinh Ba Tien	Nguyen Thanh Tung
	Nguyen Thanh Tung	Nguyen Thanh Tung
	Tran Thanh Tam	Nguyen Thanh Tung
	Nguyen Manh Hung	Nguyen Thanh Tung
	Tran Hong Quang	Tran Hong Quang
	Vuong Ngoc Quyen	Vuong Ngoc Quyen
	Le Thi Nhan	Tran Hong Quang
	Bui Thuy Vu	Tran Hong Quang

```

476 -- k) Provide the employee's full name and the names of the projects in which the employee participated, if any
477 • select concat_ws(' ', e.lastName, e.middleName, e.firstName) employeeName, p.projectName
478 from employees e
479 join projects p using(departmentID);
480

```

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	employeeName	projectName
▶	Dinh Ba Tien	San pham Z
	Dinh Ba Tien	San pham Y
	Dinh Ba Tien	San pham X
	Nguyen Thanh Tung	San pham Z
	Nguyen Thanh Tung	San pham Y
	Nguyen Thanh Tung	San pham X
	Tran Thanh Tam	San pham Z
	Tran Thanh Tam	San pham Y
	Tran Thanh Tam	San pham X
	Nguyen Manh Hung	San pham Z
	Nguyen Manh Hung	San pham Y
	Nguyen Manh Hung	San pham X
	Tran Hong Quang	Dao tao
	Tran Hong Quang	Tin hoc hoa
	Vuong Ngoc Quyen	Cap Quang
	Le Thi Nhan	Dao tao
	Le Thi Nhan	Tin hoc hoa
	Bui Thuy Vu	Dao tao
	Bui Thuy Vu	Tin hoc hoa

```

481 -- l) For each scheme, list projectName and the total number of hours worked per week of all employees attending that project.
482 • select p.projectName, sum(workingHour) total_hours
483 from projects p
484 join assignment a using(projectID)
485 group by projectID;

```

projectName	total_hours
San pham X	42.5
San pham Y	27.5
San pham Z	50
Tin hoc hoa	10
Cap Quang	25

```

487 -- m) For each department, list the name of the department (departmentName) and the average salary of the employees who work for that department.
488 • select d.departmentName, avg(e.salary) average_salary
489 from department d
490 join employees e using(departmentID)
491 group by departmentID, departmentName;

```

departmentName	average_salary
Nghien cuu	33250.0000
Dieu hanh	31000.0000
Quan ly	55000.0000

```

493 -- n) For departments with an average salary above 30,000, list the name of the department and the number of employees of that department.
494 • select d.departmentName, count(e.employeeID) employee
495 from department d
496 join employees e using(departmentID)
497 group by departmentID, departmentName
498 having avg(e.salary) > 30000;

```

departmentName	employee
Nghien cuu	4
Dieu hanh	3
Quan ly	1

```

500 -- o) Indicate the list of schemes (projectID) that has: workers with them (lastName) as 'Dinh' or ,
501 -- whose head of department presides over the scheme with them (lastName) as 'Dinh'.

```

```

502 • select p.projectID
503 from projects p
504 join department d using(departmentID)
505 join employees e using(departmentID)
506 where e.lastName = 'Dinh' or e.managerID in (
507     select e1.employeeID
508     from employees e1
509     where e1.lastName = 'Dinh'
510 );

```

projectID
3
2
1

```

512 -- p) List of employees (lastName, middleName, firstName) with more than 2 relatives.
513 • select e.lastName, e.middleName, e.firstName
514 from employees e
515 join relative r using(employeeID)
516 group by e.lastName, e.middleName, e.firstName
517 having count(*) > 2;

```

518

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	lastName	middleName	firstName
►	Dinh	Ba	Tien
	Nguyen	Thanh	Tung

```

519 -- q) List of employees (lastName, middleName, firstName) without any relatives.
520 • select e.lastName, e.middleName, e.firstName
521 from employees e
522 join relative r using(employeeID)
523 group by e.lastName, e.middleName, e.firstName
524 having not count(*) > 0;

```

525

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	lastName	middleName	firstName
--	----------	------------	-----------

```

526 -- r) List of department heads (lastName, middleName, firstName) with at least one relative
527 • select e.lastName, e.middleName, e.firstName
528 from employees e
529 join relative r using(employeeID)
530 group by e.lastName, e.middleName, e.firstName
531 having count(*) >= 1;

```

532

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	lastName	middleName	firstName
►	Dinh	Ba	Tien
	Nguyen	Thanh	Tung
	Le	Thi	Nhan

```

533 -- s) Find the surname (lastName) of unmarried department heads.
534 • select e.lastName
535 from employees e
536 join department d on e.employeeID = d.managerID
537 where d.managerID not in (
538     select r.employeeID
539     from relative r
540     where r.relationship = 'Vo Chong'
541 );

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
lastName				
Vuong				
Tran				

```

543 -- t) Indicate the full name of the employee whose salary is above the average salary of the "Research" department.
544 • select e.lastName, e.middleName, e.firstName
545 from employees e
546 where e.salary > (
547     select avg(e1.salary)
548     from employees e1
549     join department d
550     where d.departmentName = 'Nghien cuu'
551 );

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
lastName	middleName	firstName		
Nguyen	Thanh	Tung		
Nguyen	Manh	Hung		
Vuong	Ngoc	Quyen		
Le	Thi	Nhan		

```

553 -- u) Indicate the name of the department and the full name of the head of the department with the largest number of employees.
554 • select d.departmentName, concat_ws(' ', e.lastName, e.middleName, e.firstName) fullname
555 from department d
556 join employees e on d.managerID = e.employeeID
557 join employees e1 on d.departmentID = e1.departmentID
558 group by d.departmentID
559 order by count(*) desc
560 limit 1;

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
departmentName	fullname				
Nghien cuu	Nguyen Thanh Tung				

```

562 -- v) Find the full names (lastName, middleName, firstName) and addresses (Address) of employees
563 -- who work on a project in 'HCMC' but the department they belong to is not located in 'HCMC'
564 • select concat_ws(' ', e.lastName, e.middleName, e.firstName) fullname, e.address
565 from employees e
566 join projects p using(departmentID)
567 where (p.projectAddress like '%HCM%') and (e.address not like '%HCM%');
568

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
fullname	address			
Tran Thanh Tam	543 Mai Thi Luu, Ba Dinh, Ha Noi			
Nguyen Manh Hung	975 Le Lai, P3, Vung Tau			
Vuong Ngoc Quyen	450 Trung Vuong, My Tho, TG			

```

570 -- w) In general verse 16, find the names and addresses of employees who work on a scheme in a city
571 -- but the department to which they belong is not located in that city
572 • drop procedure if exists find_employees;
573 delimiter //
574 • create procedure find_employees()
575 begin
576     select concat_ws(' ', e.lastName, e.middleName, e.firstName) fullname, e.address
577     from employees e
578     join projects p using(departmentID)
579     where p.projectAddress <> e.address
580     group by e.employeeID;
581 end //
582 delimiter ;
583
584 • call find_employees();

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
fullname	address			
Dinh Ba Tien	731 Tran Hung Dao, Q1, TPHCM			
Nguyen Thanh Tung	638 Nguyen Van Cu, Q5, TPHCM			
Tran Thanh Tam	543 Mai Thi Luu, Ba Dinh, Ha Noi			
Nguyen Manh Hung	975 Le Lai, P3, Vung Tau			
Tran Hong Quang	980 Le Hong Phong, Vung Tau			
Vuong Ngoc Quyen	450 Trung Vuong, My Tho, TG			
Le Thi Nhan	291 Ho Van Hue, Q.PN, TPHCM			
Bui Thuy Vu	332 Nguyen Thai Hoc, Quy Nhon			